NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 11: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Linear regression is generally not recommended for prediction outside the range of the observed data because this involves _____.

A. interpolation.
B. regularization.
C. extrapolation.
D. normalization.

Correct Answer: (C)

Q2. Consider the following statements regarding a Maximum Margin Classifier.
I. The optimal hyperplane is the one that maximizes the margin between the two classes.
II. The linear classifier defined by the maximum-margin hyperplane is called a maximum-margin classifier.
Choose the correct option.

A. Both Statements I and II
B. Only Statement II
C. Only Statement I
D. Neither Statement I nor II

Correct Answer: (A)

Q3. Support Vector Machines (SVMs) formalize binary classification as the task of finding the maximum-margin hyperplane — the linear separator that maximizes the distance to the nearest data point on each side.
For the hard-margin formulation, which one of the following best describes a support vector?

A. Any training data point.
B. Point farthest from the hyperplane.
C. Point closest to the hyperplane.
D. Mean vector of each class.

Correct Answer: (C)

Q4. SVMs can solve non-linear classification problems because both learning and classification rely only on dot products, which can be replaced by kernel functions.
Consider the following statements.
I. Learning depends only on the dot products between pairs of training samples.

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II. Classification depends only on the dot products between the unknown sample and the training samples.

A. Only Statement I
B. Only Statement II
C. Both Statements I and II
D. Neither Statement I nor II

Correct Answer: (C)

Q5. The SVM dual formulation enables nonlinear classification through the kernel trick because:
I. The dual formulation depends on training examples through inner products.
II. A kernel function can replace these inner products to implicitly operate in a higher-dimensional feature space.
A. Statement I only.
B. Statement II only.
C. Both Statement I and II.
D. Neither Statement I nor II.

Correct Answer: (C)

Q6. Clustering is a technique for finding similarity groups in data, called clusters. Identify the correct statement(s) vis-à-vis clustering.
I. Clustering is an unsupervised learning task, where class labels are not specified and only the feature vectors representing objects/instances/records are known.
II. A cluster is a collection of data items which are "similar" between them, and "dissimilar" to data items in other clusters.

A. Only Statement I.
B. Only Statement II
C. Both Statement I and II
D. Neither Statement I nor II

Correct Answer: (C)

Q7. Inter-cluster separation measures how far apart different clusters are from one another and is a reflection of _____.

A. compactness
B. similarity
C. isolation
D. cohesion

Correct Answer: (C)

Q8. Clustering is difficult to evaluate, but it is widely used in practice. Which of the following is an example of an internal cluster evaluation measure?

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A. Compactness
B. Ground-truth comparison
C. Classification accuracy
D. Alternative-solution comparison

Correct Answer: (A)

For Q9 – Q10.

Linear regression fits a line $Y = f(X)$ to a set of data points by minimizing the sum of squared vertical distances between the observed values and the fitted line. The strength of the resulting linear relationship is quantified using the correlation coefficient r and the coefficient of determination r^2.

Q9. Suppose a regression model yield $r = 0.7$.
Which of the following correctly interprets this value?

A. There is a weak negative linear relationship between X and Y.
B. There is a strong positive linear relationship between X and Y.
C. There is a strong negative linear relationship between X and Y.
D. There is no linear relationship between X and Y.

Correct Answer: (B)

Q10. If $r = -0.7$, then r^2 and its correct interpretation are:
A. 0.49; weak relationship between X and Y.
B. 0.49; explains 70% variation in Y.
C. 0.49; explains 49% variation in Y.
D. 0.49; strong relationship between X and Y.

Correct Answer: (C)

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